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NITTE (Deemed to be University)

(Established under Sec.(3) of UGC Act, 1956)

First Semester B.Tech (CBCS) Degree Examinations, Nov. - Dec. 2025**25CSE103 – PROBLEM SOLVING THROUGH PROGRAMMING**

(For ACT, AER, BTY, CIV, ECE, EEE, MEC, VDT, AID, AIM, CYB, CCE, CSE, CBS, ISE, RAI)

Duration: 3 Hours]

[Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **One full** question from each Unit.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. The process of correcting errors in a program is called
A) Debugging B) Testing
C) Coding D) Linking
2. What is the final step of the program development cycle
A) Design B) Maintenance
C) Debugging D) Implementation
3. Computer program that converts assembly language to machine language is called
A) Compiler B) Linker
C) Assembler D) Translator
4. If a function in C does not return a value, what should be its return type
A) void B) float
C) int D) null
5. What is the purpose of an if statement in C?
A) To iterate over a block of code B) To declare a variable
C) To execute a block of code conditionally D) To define a function
6. What will happen if break is omitted in a switch case
A) Program stops immediately B) The next case statements will also execute
C) Compiler error D) Switch will restart from first case
7. Which of these is not valid in C?
A) if (x) B) if (x = 5)
C) if (x == 5) D) if x == 5
8. #include <stdio.h> int main() {int i; for (i = 0; i < 5; i++){if (i == 3) { break;} printf("%d ", i);} return 0;}
A) 0 1 2 3 B) 1 2 3
C) 0 1 2 D) 0 1 2 3 4
9. How do you access the last element of an array arr with n elements in C
A) arr[n]; B) arr[n-1];
C) arr[n+1]; D) arr[last];
10. To access the third element of an array named "arr", you would use:
A) arr[2] B) arr[3]
C) arr(2) D) arr(3)
11. Given a 2D array int a[2][3] = {{1, 2, 3}, {4, 5, 6}};, what does a[1][2] return?
A) 3 B) 4
C) 5 D) 6
12. How many passes will bubble sort make to sort the array {5, 1, 4, 2, 8}?
A) 3 B) 4
C) 5 D) 2

13. Which operator is used to access the address of a variable in C?
 A) * B) &
 C) @ D) %
14. What does the following code snippet do?

```
int main()
{ int arr[5] = {1, 2, 3, 4, 5}; int *ptr = arr; printf("%d", *(ptr + 2)); }
```

 A) Prints 2 B) Prints 3
 C) Prints the address of the third element D) Prints 4
15. Which of the following pointer declarations is invalid?
 A) int *ptr1; B) float *ptr2;
 C) char *ptr3; D) int ptr4*;
16. Which of the following is not a valid storage class in C?
 A) auto B) static
 C) volatile D) public
17. Which of the following is a collection of different data types?
 A) String B) Array
 C) Structure D) Files
18. Which function is used to open a file in C?
 A) fread() B) fopen()
 C) fclose() D) fwrite()
19. What does EOF stand for?
 A) End of file B) Error on file
 C) End of function D) Entry of file
20. If you want to add data to the end of an existing file without deleting it, which mode should you use?
 A) w B) a
 C) r D) r+

PART - B: DESCRIPTIVE ANSWER QUESTIONS

80 Marks

Unit – I

- | | Marks | BT* | CO* | PO* |
|---|--------------|------------|------------|------------|
| 1. a) Describe the essential steps involved in developing a program. | 4 | 2 | 1 | 1,2,3,8 |
| b) List and explain any four important features of the C programming language that make it powerful. | 6 | 2 | 1 | 1,2,3,8 |
| c) Given a = 12, b = 9, c = 2, evaluate and show the step-by-step result of the expression $x = a - b / 3 - c * 2 - 1$ | 6 | 3 | 1 | 1,2,3,8 |
| 2. a) List the rules for naming variables in C. Provide two examples each of valid and invalid variable names. For each invalid example, explain briefly why it violates the naming rules. | 4 | 2 | 1 | 1,2,3,8 |
| b) Write the output of the following C code:
<pre>#include <stdio.h> int main() { int a=5,b=10; printf("\n%d", a<=2); printf("\n%d", a!=b); printf("\n%d", a<<2); printf("\n%d", a>>2); printf("\n%d", a++); printf("\n%d", --a); return 0; }</pre> | 6 | 2 | 1 | 1,2,3,8 |
| c) Differentiate between pre-increment and post-increment operators in C. Illustrate their behavior with suitable code examples and show the output for each case. | 6 | 3 | 1 | 1,2,3,8 |

Unit – II

3.	a)	Compare the behavior of the while and do-while loops.	4	2	2	1,2,3,8
	b)	Write a C program to reverse a number using while loop.	6	2	2	1,2,3,8
	c)	Demonstrate your understanding of the switch-case control structure in C by writing a program to check whether a given alphabet is a vowel or a consonant.	6	3	2	1,2,3,8
4.	a)	Explain the usage of break and continue with suitable examples.	4	2	2	1,2,3,8
	b)	Interpret any two unformatted statements in C with a suitable example program.	6	2	2	1,2,3,8
	c)	Write a C program using the else-if ladder statement to calculate the electricity bill for a user based on the units consumed as per the following criteria: - Up to 50 units: ₹3.00 per unit 51 to 150 units: ₹5.00 per unit 151 to 250 units: ₹7.00 per unit - Above 250 units: ₹10.00 per unit The program should accept the total number of units as input and display the total bill amount.	6	3	2	1,2,3,8

Unit – III

5.	a)	How do you declare and initialize a 2-D array explain with suitable example.	4	2	3	1,2,3,8
	b)	Distinguish reading string using scanf() and gets() functions	4	2	3	1,2,3,8
	c)	Write a C program using a one-dimensional array to manage the salaries of 5 employees. Your program should perform the following tasks: - Accept and store the salaries of 5 employees in an array. - Display all the salaries entered by the user. Calculate and display the average salary of the employees.	8	3	3	1,2,3,8
6.	a)	Explain the following string manipulation functions with programming example. strcat(), strcmp()	4	2	3	1,2,3,8
	b)	Write a program to find transpose of a matrix.	4	2	3	1,2,3,8
	c)	Write a C program using a one-dimensional array to store the prices of 8 products in a store. Your program should: - Accept and store the prices of 8 products. - Display all the prices entered by the user. - Calculate and display the total inventory value (sum of all product prices). - Find and display the most expensive product prices.	8	3	3	1,2,3,8

Unit – IV

7.	a)	Write a C program using pointers to perform the following: - Display the address of a variable - Display the value stored at the address	5	2	4	1,2,3,8
	b)	What are actual parameters and formal parameters? Illustrate with example.	5	2	4	1,2,3,8
	c)	Write a program in C using functions to swap two numbers using call by reference concept.	6	3	4	1,2,3,8

8. a)	Illustrate the concept of 'scope of a variable' in c with an example program.	5	2	4	1,2,3,8
b)	What is the difference between call by value and call by reference?	5	2	4	1,2,3,8
c)	Write a recursive function to calculate the factorial of a number.	6	3	4	1,2,3,8

Unit – V

9. a)	Describe, what is a Structure? Explain the syntax of a structure with an example.	5	2	5	1,2,3,8
b)	Write a C program using command-line arguments to count and display the total number of words passed to the program.	5	2	5	1,2,3,8
c)	Create a structure called Employee with the fields: Name, Department and Experience, input and print the details of two employees using array of structures.	6	3	5	1,2,3,8
10. a)	Write the general syntax of fopen() and fclose() functions.	5	2	5	1,2,3,8
b)	An event registration system takes inputs from the command line: Participant Name, Age, and Event Name. Write a C program to display the details.	5	3	5	1,2,3,8
c)	Write a program to store and display details (name, age, marks) of 5 students using an array of structures.	6	2	5	1,2,3,8

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
